28. Find the Boundary of the Image using Convolution Kernel

AIM

To find the boundary or edges of an image using a convolution kernel.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Define a convolution edge-detection kernel.
4. Apply the kernel using cv2.filter2D().
5. Display the boundary image.

PROGRAM

```
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Boundary detection kernel
kernel = np.array([
    [-1, -1, -1],
    [-1,  8, -1],
    [-1, -1, -1]
])

# Apply convolution
boundary = cv2.filter2D(gray, -1, kernel)
```

Display

cv2.imshow("Original Image", img)

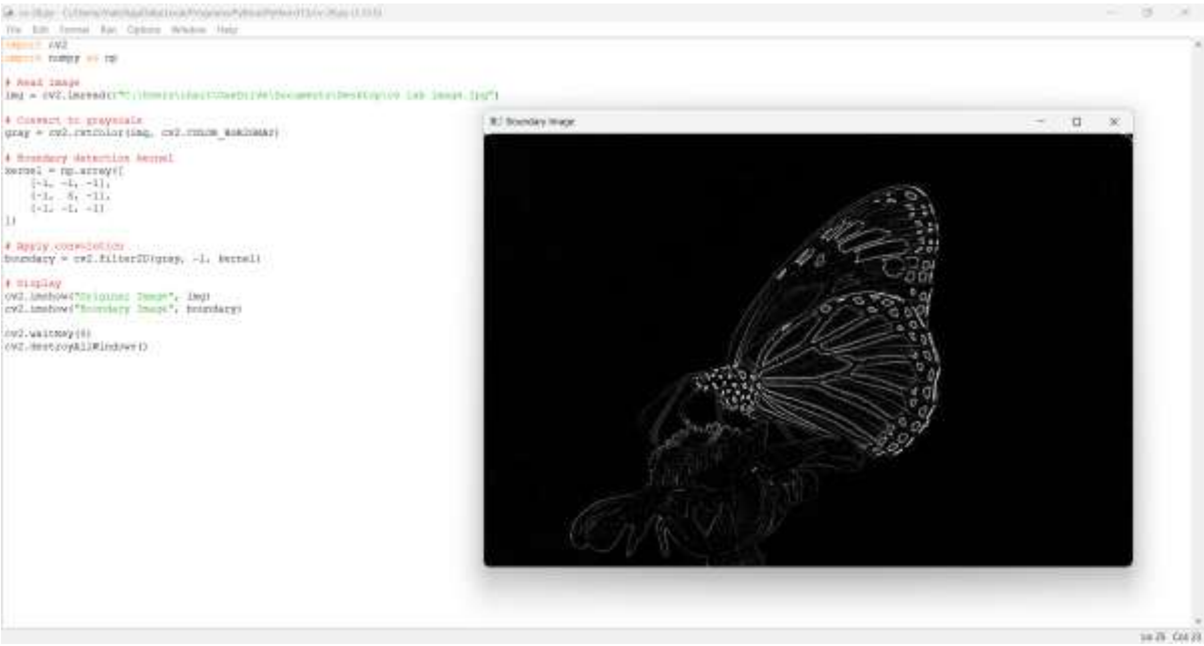
cv2.imshow("Boundary Image", boundary)

cv2.waitKey(0)

cv2.destroyAllWindows()

OUTPUT

The boundaries and edges present in the image are highlighted.



Explanation

A convolution kernel is moved over the image to detect sudden changes in intensity. The given kernel highlights boundaries and edges by comparing a pixel with its neighboring pixels.